

AWS Cloud Capstone Project: 3-Tier Architecture

Security Model: Full Isolation

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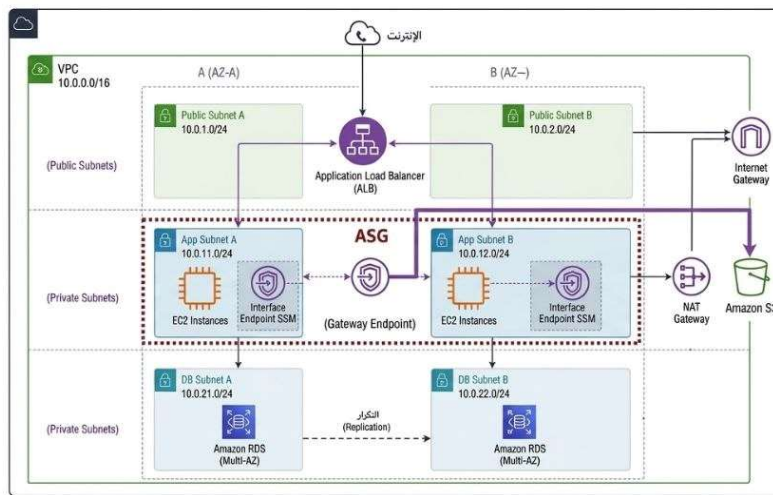
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1.Executive Summary & System Architecture Overview

This technical report documents the complete design, deployment, and verification of a production-grade 3-Tier cloud application architecture built from scratch on Amazon Web Services.

The primary objective of this architecture is to achieve maximum security, high availability across multiple Availability Zones, and dynamic scalability under variable traffic loads while maintaining zero public exposure for backend application and database layers.



Key Architectural Pillars:

- **High Availability:** Application and Database tiers are redundantly deployed across two Availability Zones ([us-east-1a](#) and [us-east-1b](#)).
- **Complete Network Isolation:** EC2 application servers and the Amazon RDS MySQL instance reside in private subnets with strictly no public IPv4 addresses assigned. Public internet ingress is restricted exclusively to the Application Load Balancer ([capstone-2026](#)).
- **Secure AWS Service Access:** All AWS service communications are routed internally via VPC Endpoints. Outbound internet access for package deployment is strictly one-way through a single NAT Gateway.
- **Zero Static Credentials:** Compute instances rely entirely on IAM Roles ([Capstone-EC2-Role](#)) with temporary credentials supplied by IMDSv2.
- **Automated Elasticity:** A dynamic Auto Scaling Group ([Capstone-ASG](#)) automatically provisions and terminates EC2 instances based on real-time CPU utilization metrics.

2. Network Infrastructure & Security Boundaries Construction

2.1 Virtual Private Cloud (VPC) & Subnet Architecture

A custom VPC (**Capstone-VPC**) was provisioned with a CIDR block of **10.0.0.0/16**. DNS Resolution and DNS Hostnames were explicitly enabled to support Private DNS resolution for Interface VPC Endpoints and RDS endpoint resolution.

Subnet Layout:

Subnet Name	Type	Availability Zone	CIDR Block	Purpose / Assigned Resources
Capstone-Public-A	Public	us-east-1a	10.0.1.0/24	Ingress ALB Node, NAT Gateway
Capstone-Public-B	Public	us-east-1b	10.0.2.0/24	Ingress ALB Node
Capstone-App-A	Private	us-east-1a	10.0.11.0/24	EC2 App Instance, SSM Endpoints
Capstone-App-B	Private	us-east-1b	10.0.12.0/24	EC2 App Instance, SSM Endpoints
Capstone-DB-A	Private	us-east-1a	10.0.21.0/24	Primary RDS Instance
Capstone-DB-B	Private	us-east-1b	10.0.22.0/24	Multi-AZ Secondary RDS Instance

Subnets (12) Info					Last updated 3 minutes ago	Actions	Create subnet
Find subnets by attribute or tag							
☐	Name	Subnet ID	State	VPC			
☐	Capstone-Public-B	subnet-04f8082df370ca4c3	Available	vpc-0071b65a223ad2928 Capstone-VPC			
☐	Capstone-Public-A	subnet-0001fdbdee12cd68b	Available	vpc-0071b65a223ad2928 Capstone-VPC			
☐	Capstone-DB-B	subnet-0fe7a850897ba6c4d	Available	vpc-0071b65a223ad2928 Capstone-VPC			
☐	Capstone-DB-A	subnet-08b573a347056d2af	Available	vpc-0071b65a223ad2928 Capstone-VPC			
☐	Capstone-App-B	subnet-0ba1c213062c23083	Available	vpc-0071b65a223ad2928 Capstone-VPC			
☐	Capstone-App-A	subnet-0900c91f0da7da3ea	Available	vpc-0071b65a223ad2928 Capstone-VPC			

2.2 Internet Gateway, NAT Gateway & Routing Tables

- **Internet Gateway (**Capstone-IGW**)**: Attached to **Capstone-VPC** to route public traffic to/from the ALB.

- **NAT Gateway (Capstone-NAT-GW):** Deployed in **Capstone-Public-A** with an Elastic IP (**Capstone-NAT-EIP**). (We should make 2 NAT gateways but we didn't because it is a paid service and it is just an educational lab).
- **Route Tables:**
 1. **Capstone-Public-RT:** Routes **0.0.0.0/0** to **Capstone-IGW**.
 2. **Capstone-App-RT:** Routes **0.0.0.0/0** to **Capstone-NAT-GW** and S3 Prefix List to **Capstone-S3-Endpoint**.
 3. **Capstone-DB-RT:** Local routes only (**10.0.0.0/16**). Strictly no route to **0.0.0.0/0**.

NAT gateways (1/1) [Info](#)

Find NAT gateways by attribute or tag

Name	NAT gateway ID	Connectivity...	State	State message	Availat
Capstone-NAT-GW	nat-0aaf28170bc507074	Public	Available	–	Zonal

nat-0aaf28170bc507074 / Capstone-NAT-GW

arm:aws:ec2:us-east-1:740820044052:natgateway/nat-0aaf28170bc507074

VPC
vpc-0071b65a223ad2928 / Capstone-VPC

100.63.214.148

Subnet
subnet-0001fdbdee12cd68b / Capstone-Public-A

10.0.1.197

Created
Sunday, September 20, 2026 at 23:15:20 GMT+3

eni-02fa661a4668737e6

Deleted
–

Internet gateways (1/2)

Find internet gateways by attribute or tag

Name	Internet gateway ID	State	VPC ID
–	igw-011b70a888ec70b89	Attached	vpc-0a333e551e315f4c7
Capstone-IGW	igw-0149e8ccdb1f02139	Attached	vpc-0071b65a223ad2928 Capstone-VPC

igw-0149e8ccdb1f02139 / Capstone-IGW

[Details](#) | [Tags](#)

Details

Internet gateway ID
[igw-0149e8ccdb1f02139](#)

State
Attached

VPC ID
[vpc-0071b65a223ad2928](#) | Capstone-VPC

Owner
740820044052

☰ VPC > Your VPCs

Your VPCs

VPCs VPC encryption controls

Your VPCs (1/2) Info

Last updated 2 minutes ago [Actions](#)

Create VPC

Find VPCs by attribute or tag

< 1 >

Name

Capstone-VPC

resource map Info

Subnets (6)

Subnets within this VPC

us-east-1a

- Capstone-Public-A
- Capstone-DB-A
- Capstone-App-A

us-east-1b

- Capstone-Public-B
- Capstone-DB-B
- Capstone-App-B

Route tables (4)

Route network traffic to resources

- Capstone-Public-RT
- Capstone-DB-RT
- Capstone-App-RT
- rtb-0007a731eccf2e7f3

Network Connections (2)

Connections to other VPCs

- Capstone-IGW
- Capstone-NAT

Route ID	Destination	Target	Status	Propagated	Route Origin
Capstone-DB-RT	10.0.0.0/16	vpce-00b688a881394b...	Active	No	Create Route
Capstone-App-RT	10.0.0.0/16	nat-0aaf28170bc507074	Active	No	Create Route
rtb-04cd78eb204590d58	10.0.0.0/16	local	Active	No	Create Route Table

rtb-00a9d6d90c3d27f1b / Capstone-App-RT

Destination	Target	Status	Propagated	Route Origin
pl-63a5400a	vpce-00b688a881394b...	Active	No	Create Route
0.0.0.0/0	nat-0aaf28170bc507074	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

3. Security Groups Chain & Isolation Principles

Security Groups were configured as a tightly chained defense perimeter where each tier explicitly references the Security Group ID of the preceding tier rather than IP ranges:

- Capstone-ALB-SG (Load Balancer):** Inbound HTTP (80) from **0.0.0.0/0**.
- Capstone-EC2-SG (Application Server):** Inbound HTTP (80) sourced exclusively from **Capstone-ALB-SG**.
- Capstone-RDS-SG (Database Layer):** Inbound MySQL (3306) sourced exclusively from **Capstone-EC2-SG**.

Capstone-Endpoint-SG (VPC Endpoints): Inbound HTTPS (443) sourced exclusively from **Capstone-EC2-SG**. Security Groups are configured using Source Security Group IDs rather than fixed CIDR blocks.

This dynamic chaining guarantees that newly auto-scaled EC2 instances inherit strict perimeter controls automatically without requiring static IP configuration. Port 22 (SSH) is completely disabled across all tiers to eliminate brute-force attack vectors.

Security Groups (6) Info					Actions	Export security groups to CSV	Create security group
Find security groups by attribute or tag					< 1 >		
☐	Name ▲	Security group ID ▼	Security group name ▼	VPC ID			
☐	Capstone-ALB-SG	sg-021187c78d966c54c	Capstone-ALB-SG	vpc-0071b65a223ad2928			
☐	Capstone-EC2-SG	sg-01bdc1e4e6391fbb	Capstone-EC2-SG	vpc-0071b65a223ad2928			
☐	Capstone-Endpoint SG	sg-055dbab45d6f67871	Capstone-Endpoint SG	vpc-0071b65a223ad2928			
☐	Capstone-RDS-SG	sg-082a9d02aa75267ef	Capstone-RDS-SG	vpc-0071b65a223ad2928			

4. Database Layer & Private AWS Service Integration

4.1 RDS MySQL Instance Configuration

- **Subnet Group:** `capstone-db-subnet-group` containing `Capstone-DB-A` and `Capstone-DB-B`.
- **Engine & DB Identifier:** MySQL Community Edition (`capstone-db`), Database Name: `capstonedb`.
- **Public Access:** Disabled (`Publicly accessible:` No).
- **Security Group:** `Capstone-RDS-SG`.

[Aurora and RDS](#) > [Subnet groups](#) > `capstone-db-subnet-group`

Subnet group details

VPC ID
[vpc-0071b65a223ad2928](#)

ARN
arn:aws:rds:us-east-1:740820044052:subgrp:capstone-db-subnet-group

Supported network types
IPv4

Description
DBs subnet group

Subnets (2)

Availability zone	Subnet name	Subnet ID	CIDR block
us-east-1b	Capstone-DB-B	subnet-0fe7a850897ba6c4d	10.0.22.0/24
us-east-1a	Capstone-DB-A	subnet-08b573a347056d2af	10.0.21.0/24

capstone-db

[Refresh](#)
[Alerts](#)
[Modify](#)
[Actions](#)

Summary

DB identifier

capstone-db

Status

Available

CPU

4.28%

Recommendations

Role

Instance

Current activity

0 Connections

Engine

MySQL Community

Region & AZ

us-east-1b

4.2 S3 Gateway Endpoint & SSM Interface Endpoints

- **S3 Gateway Endpoint (Capstone-S3-Endpoint):** Attached to **Capstone-App-RT** to allow private S3 operations.

Systems Manager Interface Endpoints: Three Interface Endpoints were provisioned with Private DNS enabled (**ssm**, **ssmmessages**, **ec2messages**). Since a single NAT Gateway was deployed in AZ-A to allow outbound internet access (e.g., package installation via **dnf**) for private EC2 instances without assigning public IPv4 addresses. To optimize security and prevent traffic from leaving the AWS private backbone, AWS service calls (S3 & SSM) bypass the NAT Gateway entirely and route through dedicated Gateway and Interface VPC Endpoints.

Endpoints (1/1) [Info](#) [Actions](#)

☒	Name	VPC endpoint ID	Endpoint type	Status
☒	Capstone-S3-Endpoint	vpce-00b688a881394bff3	Gateway	Available

vpce-00b688a881394bff3 / Capstone-S3-Endpoint

Endpoint ID
 [vpce-00b688a881394bff3](#)

VPC ID
[vpc-0071b65a223ad2928](#)
(Capstone-VPC)

DNS record ID type

Status
Available

Status message
-

IP address type
ipv4

Creation time
Sunday, September 20, 2026 at 23:58:09 GMT+3

Service name
 [com.amazonaws.us-east-1.s3](#)

Service region

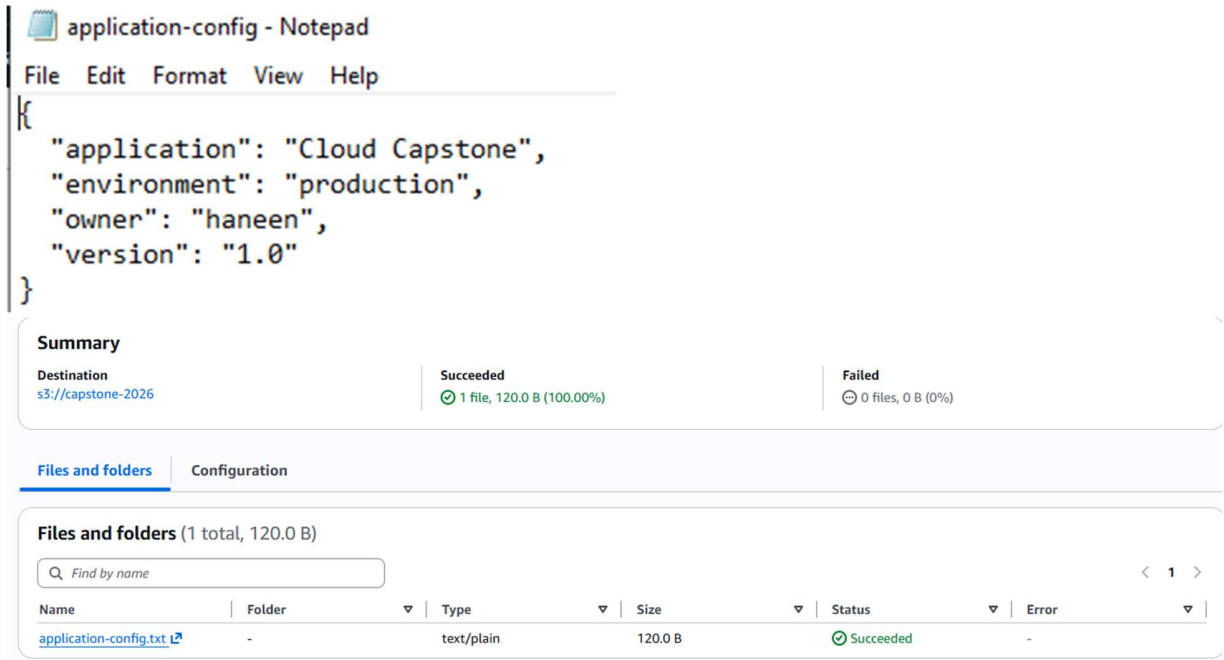
Endpoint type
Gateway

Private DNS na
No

Private DNS pr
-

Endpoints (4) [Info](#) [Actions](#) [Create](#)

☐	Name	VPC endpoint ID	Endpoint type	Status
☐	Capstone-S3-Endpoint	vpce-00b688a881394bff3	Gateway	Available
☐	Capstone-SSM-Endpoint	vpce-022404ac7e496c9cd	Interface	Available
☐	Capstone-SSMmessages-Endpoint	vpce-0fce4c56ef248b1b6	Interface	Available
☐	Capstone-EC2messages-Endpoint	vpce-0a3c6fcdb32cfa619	Interface	Available



5. Compute Layer, Launch Template & Load Balancing

5.1 IAM Role & Launch Template

- **IAM Role (Capstone-EC2-Role):** Attached policies: [AmazonSSMManagedInstanceCore](#), [AmazonS3ReadOnlyAccess](#), [CloudWatchAgentServerPolicy](#).
- **Launch Template (Capstone-LT):** Amazon Linux 2023, Instance Type [t3.micro](#), IAM Instance Profile attached. Application owner set to [haneen](#).
- Zero static credentials or access keys reside within the compute environment. EC2 instances leverage IAM Instance Profiles with IMDSv2 to acquire short-lived credentials, while remote administration is governed seamlessly via Systems Manager Session Manager.

5.2 Target Group & Application Load Balancer

- **Target Group (Capstone-TG):** Target Type: Instances, Protocol/Port: HTTP 80, Health Check Path: [/health.html](#).
- **Application Load Balancer (capstone-2026):** Internet-facing, deployed across [Capstone-Public-A](#) and [Capstone-Public-B](#) with [Capstone-ALB-SG](#).

5.3 Auto Scaling Group (Capstone-ASG)

- Capacity Settings: Desired: 2, Min: 2, Max: 4.

- Subnets: [Capstone-App-A](#) and [Capstone-App-B](#).

Permissions policies (3) [Info](#)

You can attach up to 20 managed policies.

Filter by Type

All types

☐	Policy name ↗	Type
☐	AmazonS3ReadOnlyAccess	AWS managed
☐	AmazonSSMManagedInstance...	AWS managed
☐	CloudWatchAgentServerPolicy	AWS managed

[IAM](#) > Roles

Roles (1/9) [Info](#)

An IAM role is an identity you can create that has specific permissions with credentials you trust.

☐	Role name	Trusted entities
☐	AWSServiceRoleForVPCTransitGateway	AWS Service: transitgat
☒	Capstone-EC2-Role	AWS Service: ec2

Capstone-TG [Actions](#)

Details

arn:aws:elasticloadbalancing:us-east-1:740820044052:targetgroup/Capstone-TG/c9e732da8be445b1

Target type

Instance

IP address type

IPv4

Protocol : Port

HTTP: 80

Load balancer

[None associated](#)

Protocol version

HTTP1

VPC

[vpc-0071b65a223ad2928](#)

0

Total targets

0

Healthy

0 Anomalous

0

Unhealthy

0

Unused

0

Initial

0

Draining

Capstone-ALB

Actions

▼ Details

Load balancer type

Application

Status

Active

VPC

[vpc-0071b65a223ad2928](#)

Load balancer IP address type

IPv4

Scheme

Internet-facing

Hosted zone

Z35SXDOTRQ7X7K

Availability Zones

[subnet-0001fdbdee12cd68b](#)
us-east-1a (use1-az1)
[subnet-04f8082df370ca4c3](#)
us-east-1b (use1-az2)

Date created

September 21, 2026, 01:38 (UTC+03:00)

Load balancer ARN

[arn:aws:elasticloadbalancing:us-east-1:740820044052:loadbalancer/app/Capstone-ALB/b0b5e631f6d91851](#)

DNS name Info

[Capstone-ALB-259804992.us-east-1.elb.amazonaws.com](#) (A Record)

Auto Scaling groups (1/1)

Info

Last updated less than a minute ago

Launch configurations

Launch templates

Actions

Create Auto Scaling group

Q Search your Auto Scaling groups

< 1 >

⚙

☒

Name

▼

☒

Status - new

▼

☒

Launch template/configuration

▼

☒

Instances

▼

☒

Instance health - new

▼

☒

Desired capacity

▼

☒

Capstone-ASG

At desired capacity

Capstone-LT | Version Latest

2

2/2 Healthy

2

Target groups (1/1)

Info | What's new?

Actions

Create target group

Q Filter target groups

< 1 >

⚙

☒

Name

▼

☒

ARN

▼

☒

Port

▼

☒

Protocol

▼

☒

Target type

▼

☒

Load balancer

▼

☒

Capstone-TG

arn:aws:elasticloadbalancin...

80

HTTP

Instance

Capstone-ALB

Target group: Capstone-TG

▼

Filter targets

☐

Instance ID

▼

☐

Name

▼

☐

Port

▼

☐

Zone

▼

☐

Health status

▼

☐

Health status details

▼

☐

[i-0faded1b0b3489af1](#)

-

80

us-east-1a (us...

Healthy

-

☐

[i-03450709ae8eb7bec](#)

-

80

us-east-1b (us...

Healthy

-

6. Network ACLs & Dynamic Auto Scaling Policies

6.1 Stateless Network ACL (Capstone-App-NACL)

Custom Network ACL associated with [Capstone-App-A](#) and [Capstone-App-B](#) with explicit rules for HTTP, HTTPS, MySQL, and ephemeral return traffic (1024-65535). Unlike stateful Security Groups, NACLs are stateless and inspect boundary traffic at the subnet level. Inbound Rule 120 and Outbound Rule 130 explicitly permit ephemeral ports (1024-65535) to handle session response traffic initiated by backend servers during package updates and endpoint resolution.

☒	Capstone-App-NACL	acl-01f4159c786b4fff0	2 Subnets	No	vpc-0071b65a
-------------------------------------	-------------------	-----------------------	-----------	----	--------------

acl-01f4159c786b4fff0 / Capstone-App-NACL							
Rule number	Type	Protocol	Port range	Source	Allow/Deny		
100	HTTP (80)	TCP (6)	80	10.0.0.0/16	✓	Allow	
110	HTTPS (443)	TCP (6)	443	10.0.0.0/16	✓	Allow	
120	Custom TCP	TCP (6)	1024 - 65535	0.0.0.0/0	✓	Allow	
*	All traffic	All	All	0.0.0.0/0	✗	Deny	

acl-01f4159c786b4fff0 / Capstone-App-NACL							
Rule number	Type	Protocol	Port range	Destination	Allow/Deny		
100	HTTPS (443)	TCP (6)	443	0.0.0.0/0	✓	Allow	
110	HTTP (80)	TCP (6)	80	0.0.0.0/0	✓	Allow	
120	MySQL/Aurora (33...	TCP (6)	3306	10.0.0.0/16	✓	Allow	
130	Custom TCP	TCP (6)	1024 - 65535	0.0.0.0/0	✓	Allow	
*	All traffic	All	All	0.0.0.0/0	✗	Deny	

6.2 CloudWatch Alarms & Dynamic Step Scaling Policies

1. **High CPU Alarm (Capstone-CPU-High):** Metric: `CPUUtilization` $\geq 60\%$.
2. **Low CPU Alarm (Capstone-CPU-Low):** Metric: `CPUUtilization` $\leq 20\%$.
3. **Scaling Policies:** `Capstone-Scale-Out` (+1 instance) and `Capstone-Scale-In` (-1 instance).

Capstone-ASG

Capstone-Scale-In

Policy type
Step scaling

Enabled or disabled
Enabled

Execute policy when
No alarm selected

Take the action
Remove 1 capacity units when $+\infty \leq \text{Metric name} < 0$

Capstone-Scale-Out

Policy type
Step scaling

Enabled or disabled
Enabled

Execute policy when
Capstone-CPU-High
breaches the alarm threshold: `CPUUtilization` > 60 for 2 out of 2 periods of 60 seconds for the metric dimensions:
AutoScalingGroupName = Capstone-ASG

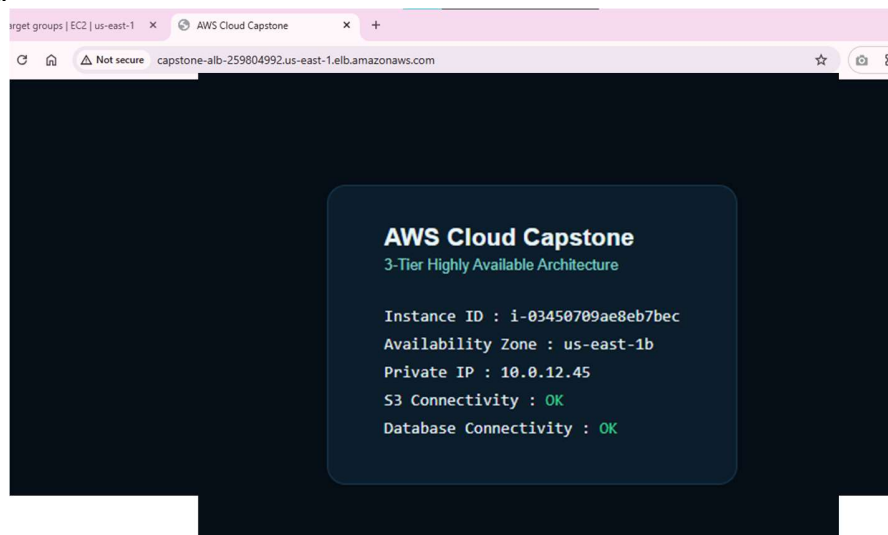
Take the action
Add 1 capacity units when $60 \leq \text{CPUUtilization} < +\infty$

Instances need
60 seconds to warm up after each step

Alarms (1/2)						
Filter alarms		Quick filters		< 1 >		
☒	Name	State	Actions	Actions muted	Last state update (UTC)	
☒	Capstone-CPU-Low	OK	Actions enabled	-	2026-09-20 23:36:12	
☐	Capstone-CPU-High	OK	Actions enabled	-	2026-09-20 23:17:01	

7. Comprehensive Verification & Test Results

1. **Test 1 - Application Ingress:** Navigated to ALB DNS URL. Page loaded showing **S3 & DB: OK**.



2. **Test 2 - Load Distribution:** Traffic balanced across instances in **us-east-1a** and **us-east-1b**.
3. **Test 3 - Database Connection:** Validated backend MySQL connectivity on port 3306.

```
[root@ip-10-0-12-130 bin]# mysql -h capstone-db.cgfy8q4m8zzw.us-east-1.rds.amazonaws.com -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 123
Server version: 8.4.9 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| capstonedbBackup |
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
5 rows in set (0.007 sec)

MySQL [(none)]>
```

4. **Test 4 - Database Isolation:** Attempted direct connection to the RDS endpoint from CloudShell; request timed out (timeout 5).

```
bash: curl: command not found
~ $ timeout 5 bash -c "cat < /dev/null > /dev/tcp/capstone-db.cgfy8q4m8zzw.us-east-1.rds.amazonaws.com/3306"
~ $
```

5. **Test 5 - Private S3 Access:** Verified `aws s3 cp` operations succeeded via Gateway Endpoint.

```
[root@ip-10-0-12-130 bin]# aws s3 ls s3://capstone-2026--region us-east-1
aws s3 cp s3://capstone-2026/application-config.txt . --region us-east-1
cat application-config.txt

Unknown options: us-east-1
download: s3://capstone-2026/application-config.txt to ./application-config.txt
{
  "application": "Cloud Capstone",
  "environment": "production",
  "owner": "haneen",
  "version": "1.0"
}[root@ip-10-0-12-130 bin]#
```

6. **Test 6 - Identity Verification:** Verification the public IP and credentials

```
[root@ip-10-0-12-130 bin]# ls -la ~/.aws 2>/dev/null || echo "No credentials file stored on the server"
No credentials file stored on the server
[root@ip-10-0-12-130 bin]#

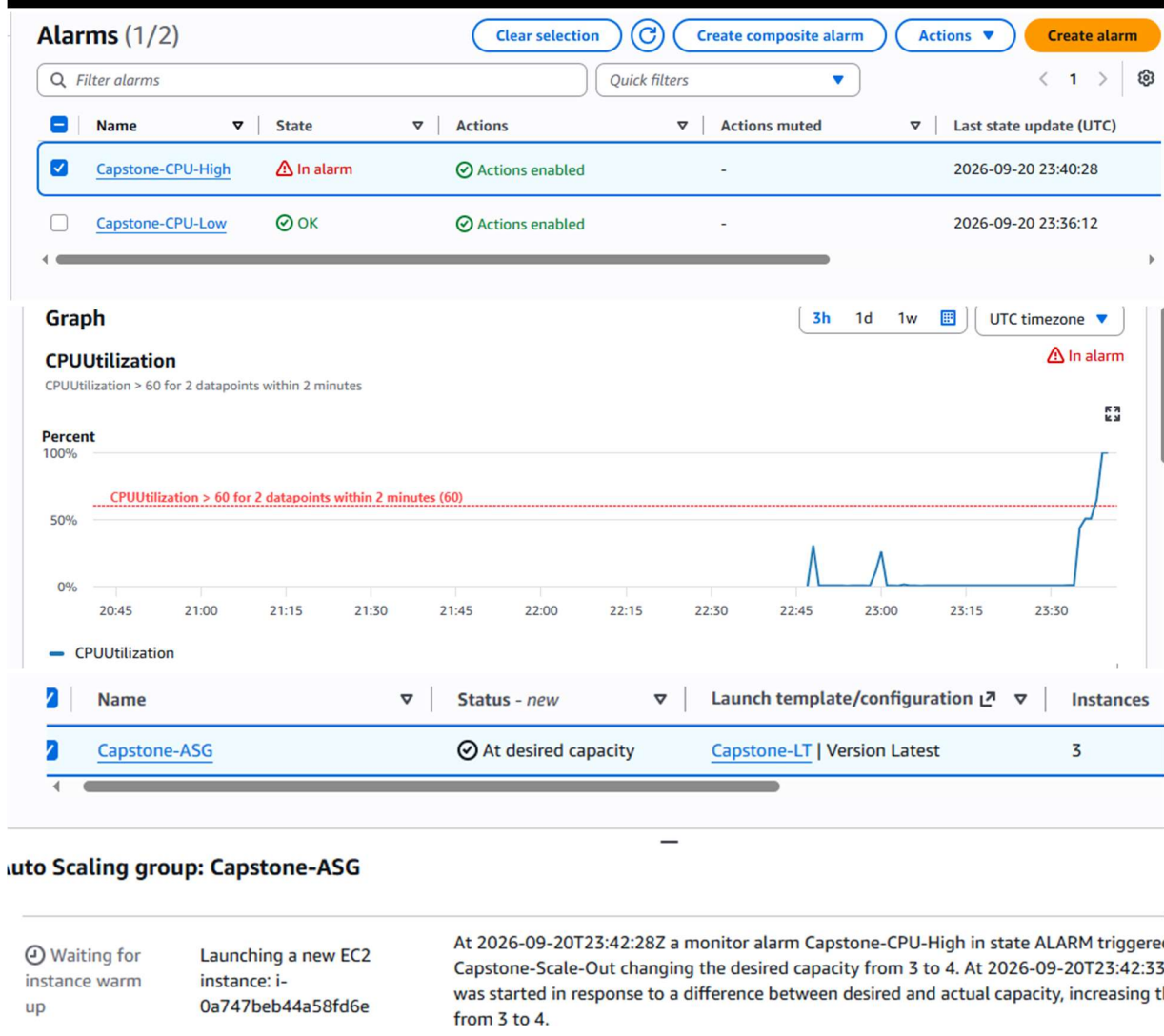
[root@ip-10-0-11-68 bin]# TOKEN=$(curl -s -X PUT "http://169.254.169.254/latest/api/token" -H "X-aws-ec2-metadata-token-ttl-seconds: 60")
curl -s -H "X-aws-ec2-metadata-token: $TOKEN" http://169.254.169.254/latest/meta-data/public-ipv4 || echo "no public IP"
<?xml version="1.0" encoding="iso-8859-1"?>
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
    "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml" xml:lang="en" lang="en">
<head>
  <title>404 - Not Found</title>
</head>
<body>
  <h1>404 - Not Found</h1>
</body>
</html>
[root@ip-10-0-11-68 bin]# TOKEN=$(curl -s -X PUT "http://169.254.169.254/latest/api/token" -H "X-aws-ec2-metadata-token-ttl-seconds: 60")
curl -fs -H "X-aws-ec2-metadata-token: $TOKEN" http://169.254.169.254/latest/meta-data/public-ipv4 || echo "no public IP - as expected"
no public IP - as expected
[root@ip-10-0-11-68 bin]#
```

7. **Test 7 - Agentless Management:** Opened terminal sessions using AWS Systems Manager Session Manager.

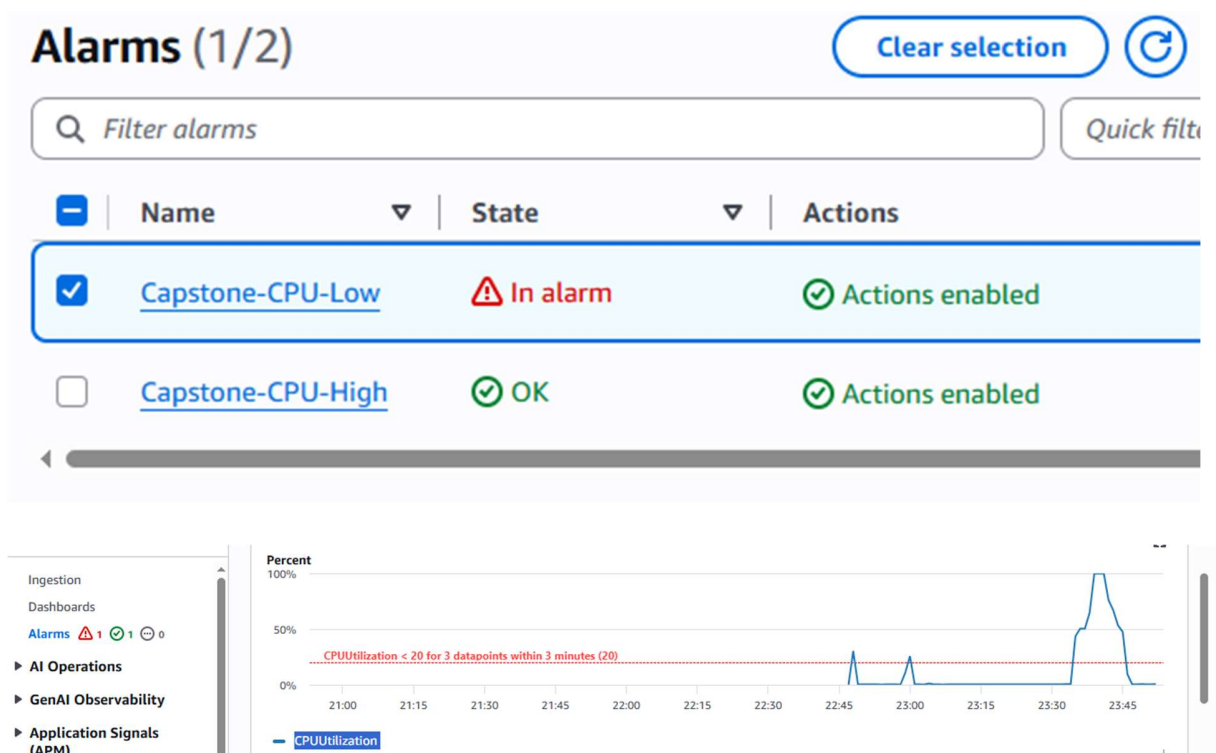
The screenshot shows the AWS Systems Manager console interface for connecting to an instance. The 'Connect to instance' dialog is open, showing four connection methods: 'Public subnet access', 'Private subnet access', 'Maximum security', and 'Direct'. The 'Maximum security' option is selected, which uses IAM authentication. Below the connection methods, the 'EC2 Instance Connect' option is also visible. The 'IAM role' section shows 'Capstone-EC2-Role' is selected. The 'Session Manager connection status' is 'Connected'. The 'SSM agent info' section shows the agent is 'Online' and the version is '3.3.4624.0'. The 'Session duration preferences' section is expanded, showing a duration of 120 minutes. The 'Manage preferences in Systems Manager' link is visible. The 'Cancel', 'Connect assist', and 'Connect' buttons are at the bottom.

8. **Test 8 - Scale-Out Under Load:** Executed multi-core CPU stress script; **Capstone-CPU-High** alarm triggered and launched a 3rd instance.

```
sh-5.2$ sudo su
[root@ip-10-0-11-68 bin]# for i in $(seq 1 $(nproc)); do
  setsid nohup bash -c 'while ;; do ;; done' >/dev/null 2>&1 &
done
[1] 27977
[2] 27978
[root@ip-10-0-11-68 bin]# uptime
 23:38:30 up 36 min,  1 user,  load average: 2.29, 1.23, 0.51
[1]-  Done                  setsid nohup bash -c 'while ;; do ;; done' > /dev/null 2>&1
[2]+  Done                  setsid nohup bash -c 'while ;; do ;; done' > /dev/null 2>&1
[root@ip-10-0-11-68 bin]#
```



9. **Test 9 - Scale-In Recovery:** Terminated stress processes; **Capstone-CPU-Low** alarm triggered and safely terminated the 3rd instance.



8. Conclusion

This Capstone project successfully demonstrates an enterprise-grade, highly available, and secure AWS infrastructure deployment.